

# ORION 1.0

Problem Statement ID: ORION-PS-03

Problem Statement Title: SylvaSense

Team ID: [ FILL THIS IN ]

Team Name: Order of One

Live demo: [ PASTE YOUR URL ]



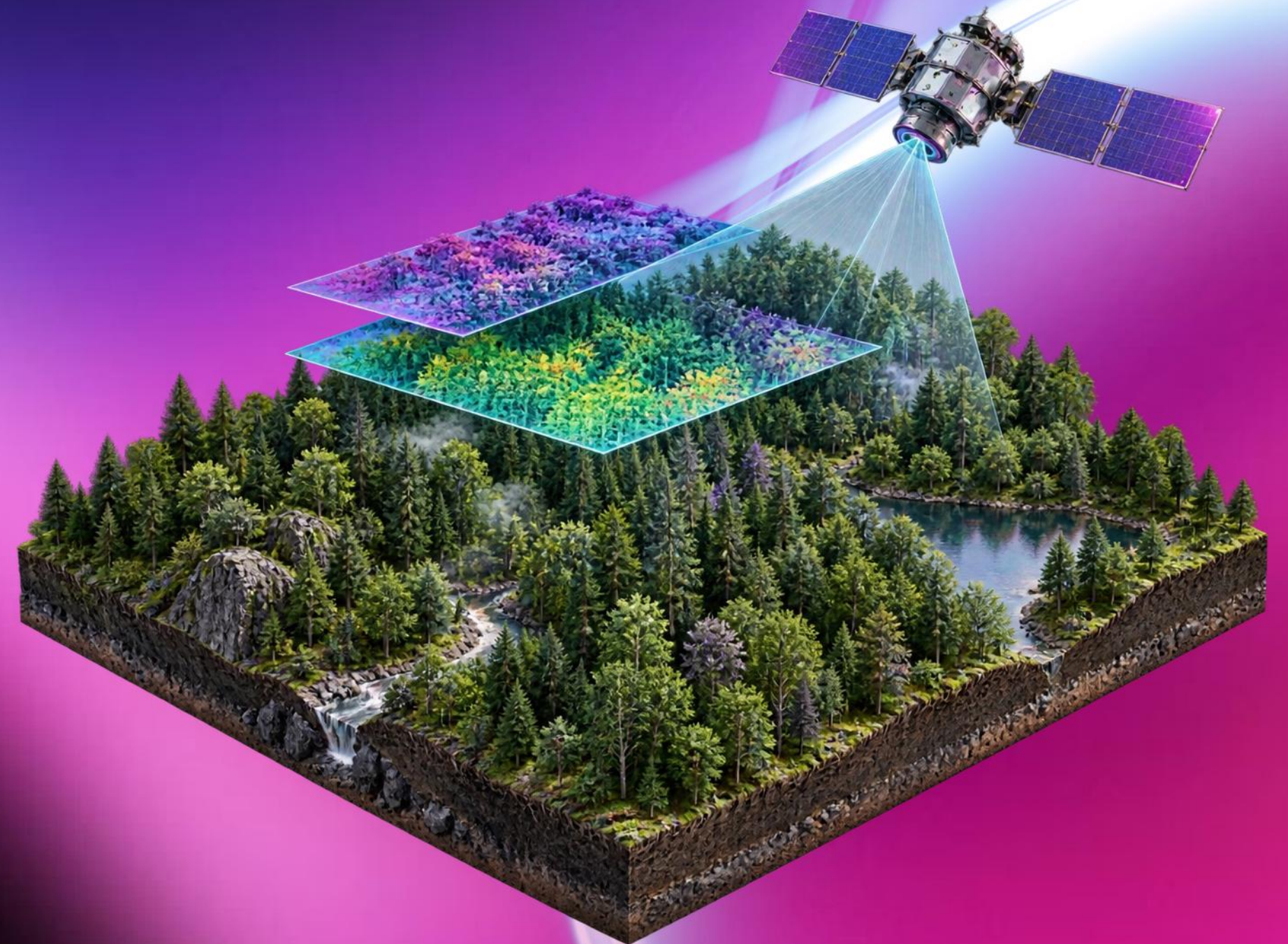
# Proposed Solution

SYLVASENSE

## From canopy to carbon.

One polygon in. Four kinds of evidence out.  
**Every number carries its uncertainty.**

- 01 Crown GeoJSON**  
Per-tree polygons, each with a confidence score
- 02 Spectral-toggle map**  
Optical, radar and structure in one frame
- 03 Biomass + validation report**  
Mg/ha with an interval that is checked
- 04 Live polygon inference**  
Draw an area; evidence returns with its sources





# Technical Approach

**Three scales. One pipeline.**

**CROWNS / FINE AERIAL IMAGERY**

**Mask2Former + Swin-S**

0.1 m NEON RGB, tiled and merged

Every mask checked against the canopy height model

**BIOMASS / OPTICAL + SAR + LIDAR**

**Gradient boosting + split conformal**

GEDI footprints are the calibration target

Coverage measured on held-out ground, not assumed

**CHANGE / MULTI-YEAR OBSERVATIONS**

**Harmonic baseline + Bayesian confirmation**

A drop must persist before it is called a detection

Below its power threshold the detector reports nothing

**$AGB = \Sigma \text{ density} \times \text{area}$     |    **Aboveground carbon =  $AGB \times CF$****

Mg/ha  $\times$  ha = Mg • CF starts at 0.47; refine by forest type [10] • Crown / biomass / change methods: [2–8]





# Feasibility & Viability

## Three gates. Already passing.

### 01 VERIFIED THE DATA

**6 sites audited on 3 independent layers**

2 of 5 candidates failed the audit and were rejected

### 02 EARNED THE MODEL CHOICE

**Spatially-blocked folds, never a random split**

Neighbouring footprints are one observation, not two

### 03 MEASURED THE TRADE-OFFS

**False positives on primary forest: 0.01–0.02%**

Coverage holds on unseen blocks — and one site fails, visibly

**41.6% → 0.01%**

A 365-day window was confirming a national park as cleared.  
Persistence is a question about time. The gate is now a span.

Measured on 6 sites, 2024–2026 • 60 automated tests • Rondônia, Brazil





# Architecture Diagram

**One polygon. Three coordinated paths.**

## INPUTS

NEON RGB  
Airborne CHM

S1 / S2 imagery  
GEDI + plot targets

Dated satellite  
observations

## QUALITY GATE

Align + tile

Match dates  
and footprints

Cloud + terrain  
quality masks

## INFERENCE

**Mask2Former**

**AGB regression**  
**Biome classifier**

**Harmonic + Bayesian**  
**Persistence test**

## OUTPUTS

Crown masks  
GeoJSON

AGB + biomes  
Uncertainty

Loss heatmap  
Carbon trend

**FastAPI job queue** → **COG + GeoJSON** → **TiTiler** → **Interactive map**

Planned architecture • Separate crown and regional scales • Biome labels: RESOLVE [9] • Every output retains date and model version

# Tech stack



## DATA

**Earth Engine · Planetary Computer · GEDI**

## PREPARE

**Rasterio · GDAL · GeoPandas · odc-stac**

## MODEL

**PyTorch · Mask2Former · split conformal**

## SERVE

**FastAPI · Redis / Celery · Docker**

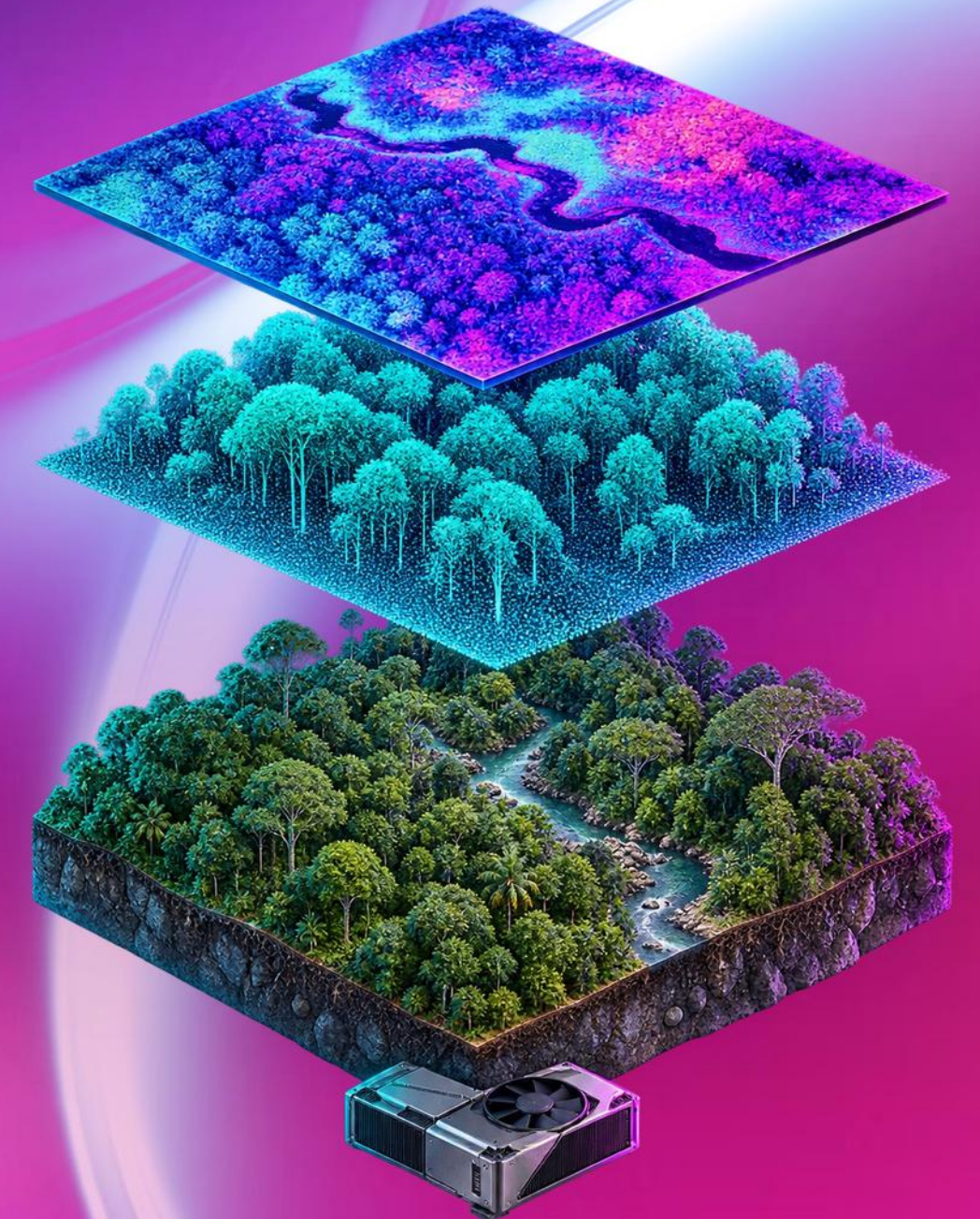
## EXPLORE

**TiTiler · Mapbox GL JS · deck.gl**

**Windowed COG reads. Cached reports. No GPU needed.**

Cold report 151–222 s · cached 3–23 ms · 7 sources, each free to fail alone

Prescribed stack, with task queues and COG tile serving added for practical inference [1,7,8]





# Impact & Benefits

**Evidence for  
every forest decision.**

## **FOREST TEAMS**

**Prioritise persistent loss alerts**

Confirmed and provisional kept apart

## **RESTORATION TEAMS**

**Track canopy and biomass trends**

Against a measured 2020 baseline

## **CARBON ANALYSTS**

**Trace stock change with its interval**

Bias-adjusted area, not pixel counts

## **COMPLIANCE TEAMS**

**Evidence for EUDR, due Dec 2026**

ESA WorldCover 2020 is the cut-off layer





# Research & References

## 01 Official scope + submission rules

ORION PS-03 · Microsoft Club SIST

## 02 Satellite observations

Sentinel-1 processing

Sentinel-2 resolution

## 03 Biomass reference targets

NASA GEDI L4A · footprints + quality

## 04 Crown-scale data

NEON 0.1 m RGB

NEON 1 m CHM

## 05 Mask2Former + crown evidence

Cheng et al., 2022

Nakada et al., 2026 preprint

## 06 Crown segmentation benchmark

Detectree2 · Ball et al., 2023

## 07 Quantile biomass regression

XGBoost · official quantile objective

## 08 Time series + raster serving

Earth Engine CCDC

TiTiler

## 09 Biome labels

RESOLVE ecoregions · Dinerstein et al., 2017

## 10 Biomass-to-carbon formulation

IPCC 2006 · Forest Land · Table 4.3